

12-Week Practice Plan — Class 4 Mathematics Olympiad

Built around the SOF IMO pattern (35 Q / 40 marks / 60 min — see [02-exam-pattern.md](#)). Three phases: build core skills topic by topic, add pure logical reasoning, then stress-test with full timed papers.

Suggested pace: **3 short sessions a week, ~30 minutes each**, not one long cram — olympiad questions reward familiarity and speed more than raw study time.

Phase 1 — Core Skills (Weeks 1–8)

One topic per week. Each week = 1 assignment of ~15 MCQs covering that topic's direct-computation + word-problem flavours (i.e. Mathematical Reasoning + Everyday Mathematics style questions together), plus a short self-check against the answer key.

Week	Topic	Folder
1	Number Sense & Numerals (place value, expanded form, ordering, rounding, Roman numerals)	assignments/01-number-sense/ ✓
2	Four Operations (+ − × ÷, multi-digit, word problems)	assignments/02-four-operations/ ✓
3	Factors, Multiples & Number Patterns	assignments/03-factors-multiples-patterns/ ✓
4	Fractions & Decimals (comparing, equivalent, simple operations)	assignments/04-fractions-decimals/ ✓
5	Measurement: Length, Weight, Capacity + Money	assignments/05-measurement-money/ ✓
6	Time & Calendar	assignments/06-time-calendar/ ✓
7	Geometry (shapes, lines, symmetry) + Perimeter & Area	assignments/07-geometry-perimeter-area/ ✓
8	Data Handling (tally marks, pictographs, bar graphs)	assignments/08-data-handling/ ✓

Weekly routine: attempt the assignment untimed first; mark it against the answer key; redo only the missed questions 2 days later to confirm the concept actually stuck.

Phase 2 — Logical Reasoning (Weeks 9–10)

Pure Section-1 skills, since these don't show up anywhere else in the school curriculum and need dedicated practice.

Week	Topic	Folder
9	Patterns, Analogy & Classification, Alphabet Test, Coding–Decoding	assignments/09-reasoning-patterns-coding/ ✓
10	Ranking Test, Direction Sense, Days & Dates, Mirror Images, Embedded Figures, Geometrical Solids	assignments/10-reasoning-direction-figures/ ✓ (24 Q — Part A text-based, Part B diagram-based with SVG figures)

Phase 3 — Achievers / HOTS + Full Papers (Weeks 11–12)

Week	Activity
11	assignments/11-achievers-hots/ ✓ — 2-and-3-step mixed problems pulling from every Phase-1 topic. Then attempt Sample Paper 1 (untimed, open-book) to find the gaps — see sample-papers/README.md .
12	Attempt Sample Papers 2–5 (Level-1), each strictly timed to 60 minutes , one every 1–2 days. After each paper, log every wrong answer in an error log (topic + why it was wrong) before moving to the next paper.
13 (if qualified)	Work through the 3 Level-2 topic assignments (assignments/13-achievers-level2/ , 14-reasoning-level2/ , 15-geometry-level2/) untimed first, to rebuild each skill at the harder difficulty. Then attempt Sample Papers 6–9 (Level-2 track), timed, once Level 1 results come in and you've qualified for Level 2 — see below.

Sample paper schedule (Phase 3)

Paper	When	Mode
Sample Paper 1	Week 11, end	Untimed, review-focused
Sample Paper 2	Week 12, Day 1	Timed (60 min)
Sample Paper 3	Week 12, Day 3	Timed (60 min)
Sample Paper 4	Week 12, Day 5	Timed (60 min)
Sample Paper 5	Week 12, Day 7	Timed (60 min) — treat as Level-1 dress rehearsal
Sample Paper 6	After qualifying Level 1	Timed (60 min), Level-2 difficulty
Sample Paper 7	1–2 days after Paper 6	Timed (60 min), Level-2 difficulty

Paper	When	Mode
Sample Paper 8	1–2 days after Paper 7	Timed (60 min), Level-2 difficulty
Sample Paper 9	1–2 days after Paper 8	Timed (60 min), Level-2 difficulty

Level-2 topic assignments (new)

Before the Level-2 sample papers, three dedicated harder topic worksheets are available, for the areas where Level 2 gets conceptually harder (not just bigger numbers):

Folder	Topic
assignments/13-achievers-level2/ 	Multi-step HOTS (working backwards, fraction-of-remainder, age/rate problems)
assignments/14-reasoning-level2/ 	Compound logical reasoning (2+ stacked clues per question)
assignments/15-geometry-level2/ 	Compound-shape perimeter & area (paths, frames, cut corners)

All 9 sample papers are built out (full 35 questions, 40 marks, answer key each) — see [sample-papers/README.md](#). Papers 1–5 are Level-1 difficulty; Papers 6–9 are a dedicated **Level-2 track** for students who've qualified Level 1 and are prepping for the actual, harder Level-2 exam.

Progress tracker

Copy this checklist and tick off as you go:

- ☐ Week 1 — Number Sense (*assignment ready*)
- ☐ Week 2 — Four Operations (*assignment ready*)
- ☐ Week 3 — Factors, Multiples & Patterns (*assignment ready*)
- ☐ Week 4 — Fractions & Decimals (*assignment ready*)
- ☐ Week 5 — Measurement & Money (*assignment ready*)
- ☐ Week 6 — Time & Calendar (*assignment ready*)
- ☐ Week 7 — Geometry, Perimeter & Area (*assignment ready*)
- ☐ Week 8 — Data Handling (*assignment ready*)
- ☐ Week 9 — Reasoning: Patterns/Analogy/Coding (*assignment ready*)
- ☐ Week 10 — Reasoning: Ranking/Direction/Calendar (*assignment ready*)
- ☐ Week 11 — Achievers/HOTS + Sample Paper 1 (*assignment + paper ready*)
- ☐ Week 12 — Sample Papers 2–5, timed (*Level-1, all ready*)

- ☐ Week 13 — Level-2 topic assignments (Achievers, Reasoning, Geometry), if qualified (*all ready*)
- ☐ Week 13 — Sample Papers 6–9, timed, if qualified for Level 2 (*Level-2 track, all ready*)

(The checkboxes above track *your* progress working through the plan, not whether the content exists — every assignment and sample paper listed is already built.)

Optional extension

Once the 12 weeks and all 6 sample papers are comfortably done (consistently $\geq 32/40$, i.e. Level-2 cutoff range), move to [assignments/12-math-kangaroo-stretch/](#)  — 15 Math Kangaroo-style puzzle questions (shape counting, mirror images, balance-scale logic, grid paths, cube nets, and classic brain-teasers like the "snail in the well") in 3 difficulty tiers, for an extra spatial-reasoning/creative-problem-solving stretch beyond anything the core plan covers. See the note in [01-syllabus.md](#) §3.